NUCLEAR CHEMISTRY 1

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Nuclear chemistry studies the nucleus (centre) of atoms. In 1909, Ernest Rutherford was the first person to come up with the idea that the nucleus of an atom was very small, dense and positively charged. Rutherford calculated that the radius of an atom would be in the order of 10^{-10} m and the radius of the nucleus would be approximately 10^{-14} m.

1 How does the size of the radius of the nucleus compare to that of the whole atom?

Shortly after this (1913) it was calculated that the protons of an individual atom made up about half of the mass of the nucleus. Hence there must be some other (unknown) particles making up the 'missing' mass. The concept of **atomic number** (Z) came into use around this time. The atomic number of a particular atom of an element shows:

- the number of electrons
- the number of protons
- the order in which the element is listed in the Periodic Table.

In 1932, James Chadwick discovered that inside atoms there were uncharged particles with a mass approximately that of a proton. We now call this particle a **neutron** and know that it is located inside the nucleus.

2 Copy and complete the table.

Particle	Mass (amu)	Electrical charge	Position
	5×10^{-4}		In shells outside the nucleus
Neutron		None	
	1.00		In the nucleus

All atoms of an element have the same atomic number (Z), but some atoms have a different atomic mass (A). This is because they can have different numbers of neutrons in the nucleus. Atoms of the same elements that have different masses are called **isotopes**. English physicist Francis Aston (1877–1945) was awarded a Nobel Prize in 1922 for his invention of the **mass spectrograph**. This machine can show the relative percentages present of different isotopes of the same element.

Figure 1 shows a simplified mass spectrometer trace for the element chlorine. Each peak shows the presence of a particular isotope. The height of the peak shows the relative abundance of each isotope.

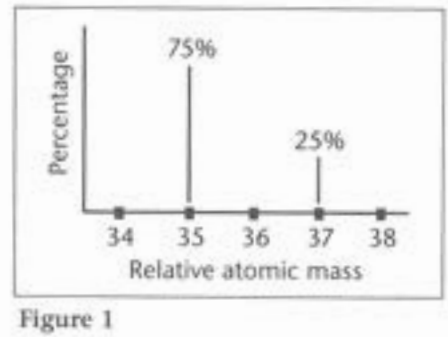


Figure 1

3 How many isotopes of chlorine are shown in figure 1?

4 What is the mass of the more common isotope of chlorine?

In nature, chlorine exists as a mixture of chlorine-35 (75%) and chlorine-37 (25%). Figure 2 shows how to calculate the average mass of naturally-occurring chlorine.

5 Use the equation shown in figure 2 to calculate the average relative atomic mass of chlorine. Don't forget to include units.

$$\frac{(75 \times 35.0) + (25 \times 37.0)}{100}$$

Figure 2

6 Copper exists in nature as two main isotopes: copper–63 (69%) and copper–65 (31%). Calculate the average relative atomic mass of copper.

7 Match each term to the correct description: *amu, isotope, mass spectrograph, neutron.*

- **a** An atomic particle that has no electrical charge and a mass approximately that of a proton
- **b** A machine that gives the relative atomic masses and abundances of atoms
- **c** An atom of an element that has different numbers of neutrons to other atoms of that element
- **d** The unit of mass used when dealing with atoms and molecules

Did you know? The neutron gets its name from the Latin *neuter* meaning 'neither' because it has neither a negative nor a positive charge.

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Radioactivity

Radioactivity is the process by which an unstable atomic nucleus is transformed into another product (element) by emitting ionising radiation.

Henri Becquerel (1852–1908) studied the properties of uranium compounds. In 1896 he found that they fogged photographic plates even when the plates were wrapped in an opaque covering. The radiation given off by uranium resembled X-rays (discovered by Roentgen in 1895) but could be deflected by electric or magnetic fields, which did not affect X-rays. Hence Becquerel's radiation must be made up of charged particles. Becquerel shared the Nobel Prize for Physics in 1903 with Pierre Curie (1859–1906) and Marie Curie (1867–1934). The term 'radioactivity' was first used by Marie Curie.

Further studies revealed that there were other naturally radioactive elements, such as thorium, radon, radium and polonium. Three main types of ionising radioactivity were identified: alpha (α), beta (β) and gamma (γ). Figure 3 summarises how radiation from radium behaves when it comes into contact with thin sheets of different materials.

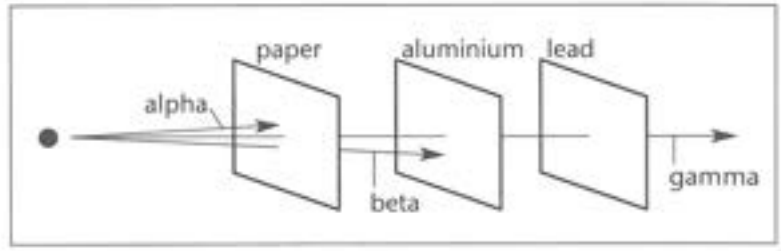


Figure 3

8 Give the name and symbol of each radioactive element mentioned so far on this page.

Figure 4 shows what happens when a beam of ionising radiation is placed in a magnetic field. A similar effect is seen if an electric charge is brought close to this radiation. Alpha particles are helium nuclei ($^4_2He^{2+}$) and they are made of two neutrons and two protons.

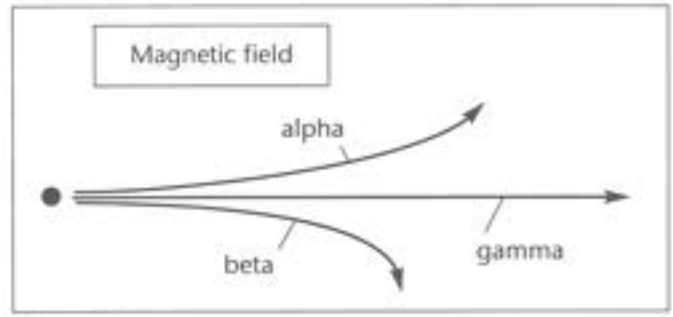


Figure 4

Alpha particles travel at a range of speeds, but around 5% of the speed of light is a typical value. (Light travels at approximately 300 000 km.s^{-1}.) The main source of alpha particles in nature is radon gas, which itself is a breakdown product of uranium.

Beta radiation is a stream of fast-moving electrons (called beta particles) emitted by a nucleus during radioactive decay. These particles have very little mass (about 5.41×10^{-4} amu) and travel close to the speed of light. **Gamma radiation** is at the high frequency (and hence high energy, short wavelength) end of the electromagnetic spectrum. It travels at the speed of light.

Problems with radiation When alpha, beta and gamma radiation collide with neutral particles they can remove electrons from these neutral particles and convert them into charged particles (ions). Exposure to ionising radiation can damage chromosomes and cause cancers and tumours. Alpha particles are relatively heavy, cause a lot of ionisation and can damage body cells. Airborne alpha radiation does not usually pose much of a hazard because it can only travel a short distance in air and is stopped by clothing. If an alpha source is taken into the body (e.g. by contaminated food or water) then it can cause severe problems. Beta particles are much less ionising than alpha particles and are much lighter but are more penetrating. However some radioisotopes that emit beta particles are soluble. Once these are in the food chain they can harm humans if ingested. Gamma radiation, because it has a very short wavelength, will pass through cells usually causing little damage unless the radiation is very intense.

9 Draw up a table to summarise the properties of radiation. Use the headings shown below.

Name	Nature of radiation	Speed	Stopped by	Ionising strength

10 Explain why alpha, beta and gamma radiation are referred to as *ionising radiation*.

11 Explain why alpha radiation can cause damage if it enters the human body.

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